

Confirmation of HLA-II associations with TB susceptibility in admixed African samples

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eLife Assessment

This **valuable** study confirms the association between the human leukocyte antigen (HLA)-II region and tuberculosis (TB) susceptibility in genetically admixed South African populations, specifically identifying a near-genome-wide significant association in the HLA-DPB1 gene, which originates from KhoeSan ancestry. The evidence supporting the association between the HLA-II region and TB susceptibility is **solid**, and the work will be of interest to those studying the genetic basis of tuberculosis susceptibility/infection resistance.

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Abstract

The International Tuberculosis Host Genetics Consortium (ITHGC) demonstrated the power of large-scale GWAS analysis across diverse ancestries in identifying tuberculosis (TB) susceptibility loci. Despite identifying a significant genetic correlate in the human leukocyte antigen (HLA)-II region, this association did not replicate in the African ancestry-specific analysis, due to small sample size and the inclusion of admixed samples. Our study aimed to build upon the findings from the ITHGC and identify TB susceptibility loci in an admixed South African cohort using the local ancestry allelic adjusted association (LAAA) model. We identified a near-genome-wide significant association (*rs3117230*, p -value = 5.292×10^{-6} , OR = 0.437, SE = 0.182) in the *HLA-DPB1* gene originating from KhoeSan ancestry. These findings extend the work of the ITHGC, underscore the need for innovative strategies in studying complex admixed populations, and confirm the role of the HLA-II region in TB susceptibility in admixed South African samples.

Introduction

Tuberculosis (TB) is a communicable disease caused by *Mycobacterium tuberculosis* (*M.tb*) (World Health Organization, 2023 [\[1\]](#)). *M.tb* infection has a wide range of clinical manifestations from asymptomatic, non-transmissible, or so-called “latent”, infections to active TB (Zaidi et al., 2023 [\[2\]](#)). Approximately 1/4 of the global population is infected with *M.tb*, but only 5-15% of infected individuals will develop active TB (Menzies et al., 2021 [\[3\]](#)). Several factors increase the risk of progressing to active TB, including co-infection with human immunodeficiency virus (HIV) and comorbidities, such as diabetes mellitus, asthma and other airway and lung diseases (Glaziou et al., 2018 [\[4\]](#)). Socio-economic factors including smoking, malnutrition, alcohol abuse, intravenous drug use, prolonged residence in a high burdened community, overcrowding, informal housing and poor sanitation also influence *M.tb* transmission and infection (Cudahy et al., 2020 [\[5\]](#); Escombe et al., 2019 [\[6\]](#); Laghari et al., 2019 [\[7\]](#); Matose et al., 2019 [\[8\]](#); Smith et al., 2023 [\[9\]](#)). Additionally, individual variability in infection and disease progression has been attributed to variation in the host genome (Schurz et al., 2024 [\[10\]](#); Caitlin Uren et al., 2021 [\[11\]](#); Verhein et al., 2018 [\[12\]](#)). Numerous genome-wide association studies (GWASs) investigating TB susceptibility have been conducted across different population groups. However, findings from these studies often do not replicate across population groups (Möller & Kinnear, 2020 [\[13\]](#); Möller et al., 2018 [\[14\]](#); Caitlin Uren et al., 2017 [\[15\]](#)). This lack of replication could be caused by small sample sizes, variation in phenotype definitions among studies, variation in linkage disequilibrium (LD) patterns across different population groups and the presence of population-specific effects (Möller & Kinnear, 2020 [\[13\]](#)). Additionally, complex LD patterns within population groups, produced by admixture, impede the detection of statistically significant loci when using traditional GWAS methods (Swart et al., 2020 [\[16\]](#)).

The International Tuberculosis Host Genetics Consortium (ITHGC) performed a meta-analysis of TB GWAS results including 14 153 TB cases and 19 536 controls of African, Asian and European ancestries (Schurz et al., 2024 [\[10\]](#)). The multi-ancestry meta-analysis identified one genome-wide significant variant (*rs28383206*) in the human leukocyte antigen (HLA)-II region ($p = 5.2 \times 10^{-9}$, OR = 0.89, 95% CI = 0.84-0.95). The association peak at the *HLA-II* locus encompassed several genes encoding crucial antigen presentation proteins (including *HLA-DR* and *HLA-DQ*). While ancestry-specific association analyses in the European and Asian cohorts also produced suggestive peaks in the HLA-II region, the African ancestry-specific association test did not yield any associations or suggestive peaks. The authors described possible reasons for the lack of associations, including the smaller sample size compared to the other ancestry-specific meta-analyses, increased genetic diversity within African individuals and population stratification produced by two admixed cohorts from the South African Coloured (SAC) population (Schurz et al., 2024 [\[10\]](#)). The SAC population (as termed in the South African census (Lehohla, 2012 [\[17\]](#))) form part of a multi-way (up to five-way) admixed population with ancestral contributions from Bantu-speaking African (~30%), KhoeSan (~30%), European (~20%), and East (~10%) and Southeast Asian (~10%) populations (Chimusa et al., 2013 [\[18\]](#)). The diverse genetic background of admixed individuals can lead to population stratification, potentially introducing confounding variables. However, the power to detect statistically significant loci in admixed populations can be improved by leveraging admixture-induced local ancestry (Swart et al., 2021 [\[19\]](#); Swart, van Eeden, et al., 2022). Since previous computational algorithms were not able to include local ancestry as a covariate for GWASs, the local ancestry allele adjusted association model (LAAA) was developed to overcome this limitation (Duan et al., 2018 [\[20\]](#)). The LAAA model identifies ancestry-specific alleles associated with the phenotype by including the minor alleles and the corresponding ancestry of the minor alleles (obtained by local ancestry inference) as covariates. The LAAA model has been successfully applied in a cohort of multi-way admixed SAC individuals to identify novel variants associated with TB susceptibility (Swart et al., 2021 [\[19\]](#); Swart, van Eeden, et al., 2022).

Our study builds upon the findings from the ITHGC (Schurz et al., 2024) and aim to resolve the challenges faced in African ancestry-specific association analysis. Here, we explore host genetic correlates of TB in a complex admixed SAC population using the LAAA model.

Methods

Data

This study included the two SAC admixed datasets from the ITHGC analysis [RSA(A) and RSA(M)] as well as four additional TB case-control datasets obtained from admixed South African population groups (Table 1). Like the SAC population, the Xhosa population are admixed with rain-forest forager and KhoeSan ancestral contributions (Choudhury et al., 2021). All datasets were collected over the past 30 years under different research projects (Daya et al., 2013; Kroon et al., 2020; Schurz et al., 2018; Smith et al., 2023; Ugarte-Gil et al., 2020) and individuals that were included in the analyses consented to the use of their data in future research regarding TB host genetics. Across all datasets, TB cases were bacteriologically confirmed (culture positive) or diagnosed by GeneXpert. Controls were healthy individuals with no previous or current history of TB disease or treatment. However, given the high prevalence of TB in South Africa [852 cases (95% CI 679-1026) per 100 00 individuals 15 years and older (Cudahy et al., 2020)], most controls have likely been exposed to *M.tb* at some point (Gallant et al., 2010). For all datasets, cases and controls were obtained from the same community and thus share similar socio-economic status and health care access.

A list of sites genotyped on the InfiniumTM H3Africa array (<https://chipinfo.h3abionet.org/browse>) were extracted from the whole-genome sequenced [RSA(Xhosa)] dataset and treated as genotype data in subsequent analyses. Quality control (QC) of raw genotype data was performed using PLINK v1.9 (Purcell et al., 2007). In all datasets, individuals were screened for sex concordance and discordant sex information was corrected based on X chromosome homozygosity estimates ($F_{\text{estimate}} < 0.2$ for females and $F_{\text{estimate}} > 0.8$ for males). In the event that sex information could not be corrected based on homozygosity estimates, individuals with missing or discordant sex information were removed. Individuals with genotype call rates less than 90% and SNPs with more than 5% missingness were removed as described previously (Swart et al., 2021). Monomorphic sites were removed. Individuals were screened for deviations in Hardy-Weinberg Equilibrium (HWE) for each SNP and sites deviating from the HWE threshold of 10^{-5} were removed. Sex chromosomes were excluded from the analysis. The genome coordinates across all datasets were checked for consistency and, if necessary, converted to GRCh37 using the UCSC liftOver tool (Kuhn et al., 2013).

Genotype datasets were pre-phased using SHAPEIT v2 (Delaneau et al., 2013) and imputed using the Positional Burrows-Wheeler Transformation (PBWT) algorithm through the Sanger Imputation Server (SIS) (Durbin, 2014). The African Genome Resource (AGR) panel (n=4 956), accessed via the SIS, was used as the reference panel for imputation (Gurdasani et al., 2015) since it has been shown that the AGR is the best reference panel for imputation of missing genotypes for samples from the SAC population (Schurz et al., 2019). Imputed data were filtered to remove sites with imputation quality INFO scores less than 0.95. Individual datasets were screened for relatedness using KING software (Manichaikul et al., 2010) and individuals up to second degree relatedness were removed. A total of 7 544 769 markers overlapped across all six datasets. This list of intersecting markers was extracted from each dataset using PLINK --extract flag. The datasets were then merged using the PLINK v1.9. After merging, all individuals missing more than 10% genotypes were removed, markers with more than 5% missing data were excluded and a HWE filter was applied to controls (threshold $<10^{-5}$). The merged dataset was screened for relatedness using KING and individuals up to second degree relatedness were subsequently removed. The

Dataset	Genotyping platform	Self-reported ethnicity	Cases/controls	Reference
RSA(A)	Affymetrix 500k	SAC	642/91	(Daya et al., 2013)
RSA(M)	MEGA array 1.1M	SAC	555/440	(Schurz et al., 2018; Swart et al., 2021)
RSA(TANDEM)	H3Africa array	SAC and Bantu-speaking African	161/133	(Swart, Uren, et al., 2022)
RSA(NCTB)	H3Africa array	SAC	49/111	(Oyageshio et al., 2023)
RSA(Worcester)	H3Africa array	SAC	61 cases	Unpublished
RSA(Xhosa)	Whole genome sequencing	IsiXhosa	44/120	Unpublished

Table 1.

Summary of the datasets included in analysis.

final merged dataset after QC and data filtering (including the removal of related individuals) consisted of 1 544 individuals (952 TB cases and 592 healthy controls). A total of 7 510 057 variants passed QC and filtering parameters.

Global ancestry inference

ADMIXTURE was used to determine the correct number of contributing ancestral proportions in our multi-way admixed population cohort (Alexander & Lange, 2011 [↗](#)). ADMIXTURE estimates the number of contributing ancestral populations (denoted by K) and population allele frequencies through cross-validation (CV). All 1 544 individuals were grouped into running groups of equal size together with 191 reference populations (Table 2 [↗](#)). Running groups were created to ensure approximately equal numbers of reference populations and admixed populations. Xhosa and SAC samples were divided into separate running groups.

Redundant SNPs were removed by PLINK through LD pruning by removing each SNP with $LD\ r^2 > 0.1$ within a 50-SNP sliding window (advanced by 10 SNPs at a time). Ancestral proportions were inferred in an unsupervised manner for $K = 3-6$ (1 iteration). The best value of K for the data was selected by choosing the K value with the lowest CV error across all running groups. Ten iterations of $K = 3$ and $K = 5$ was run for the Xhosa and SAC individuals respectively. Since it has been shown that RFMix (Maples et al., 2013 [↗](#)) outperforms ADMIXTURE in determining global ancestry proportions (C Uren et al., 2020 [↗](#)), RFMix was also used to refine inferred global ancestry proportions. Global ancestral proportions were visualised using PONG (Behr et al., 2016 [↗](#)).

Local ancestry inference

The merged dataset and the reference file (containing reference populations from Table 2 [↗](#)) were phased separately using SHAPEIT2. The local ancestry for each position in the genome was inferred using RFMix (Maples et al., 2013 [↗](#)). Default parameters were used, but the number of generations since admixture was set to 15 for the SAC individuals and 20 for the Xhosa individuals (as determined by previous studies) (Caitlin Uren et al., 2016 [↗](#)). RFMix was run with three expectation maximisation iterations and the `--reanalyse-reference` flag.

Batch effect screening and correction

Merging separate datasets generated at different timepoints and/or facilities, as we have done here, will undoubtedly introduce batch effects. Principal component analysis (PCA) is a common method used to visualise batch effects, where the first two principal components (PCs) are plotted with each sample coloured by batch, and a separation of colours is indicative of a batch effect (Nyamundanda et al., 2017 [↗](#)). However, it is difficult to differentiate between separation caused by population structure and separation caused by batch effect using PCA alone. An alternative method to detect batch effects (Chen et al., 2022 [↗](#)) involves coding case/control status by batch followed by running an association analysis testing each batch against all other batches. If any single dataset has more positive signals compared to the other datasets, then batch effects may be responsible for producing spurious results. Batch effects can be resolved by removing those SNPs which pass the genome-wide significance threshold from the merged dataset. We have adapted this batch effect correction method for application in a highly admixed cohort with complex population structure (Croock et al., 2024). Our modified method was used to remove 36 627 SNPs affected by batch effects from our merged dataset.

Local ancestry allelic adjusted association analysis

The LAAA association model was used to investigate if there are allelic, ancestry-specific or ancestry-specific allelic associations with TB susceptibility in our merged dataset. Global ancestral components inferred by RFMix, age and sex were included as covariates in the association tests. Variants with minor allele frequency (MAF) $< 1\%$ were removed to improve the stability of the association tests. Dosage files, which code the number of alleles of a specific ancestry at each locus

Table 2.

Ancestral populations included for global ancestry deconvolution.

Population	n	Source
European (British – GBR)	40	1000 Genomes (1000G) phase 3 (1000 Genomes Project Consortium et al., 2015)
East Asian (Chinese – CHB)	40	1000G phase 3
Bantu-speaking African (Yoruba – YRI)	40	1000G phase 3
Southeast Asian (Malaysian)	38	Singapore Sequencing Malay Project (SSMP) (Wong et al., 2013)
KhoeSan (Nama)	33	African Genome Variation Project (AGVP/ADRP) (Gurdasani et al., 2015)

across the genome, were compiled. Separate regression models for each ancestral contribution were fitted to investigate which ancestral contribution is associated with TB susceptibility. Details regarding the models have been described elsewhere (Swart, van Eeden, et al., 2022); but in summary, four regression models were tested to detect the source of the association signals observed:

(1) Null model or global ancestry (GA) model

The null model only includes global ancestry, sex and age covariates. This test investigates whether an additive allelic dose exerts an effect on the phenotype (without including local ancestry of the allele).

(2) Local ancestry (LA) model

This model is used in admixture mapping to identify ancestry-specific variants associated with a specific phenotype. The LA model evaluates the number of alleles of a specific ancestry at a locus and includes the corresponding marginal effect as a covariate in association analyses.

(3) Ancestry plus allelic (APA) model

The APA model simultaneously performs model (1) and (2). This model tests whether an additive allelic dose exerts an effect of the phenotype whilst adjusting for local ancestry.

(4) Local ancestry adjusted allelic (LAAA) model

The LAAA model is an extension of the APA model, which models the combination of the minor allele and ancestry of the minor allele at a specific locus and the effect this interaction has on the phenotype.

The R package *STEAM* (Significance Threshold Estimation for Admixture Mapping) (Grinde et al., 2019) was used to determine the genome-wide significance threshold given the global ancestral proportions of each individual and the number of generations since admixture ($g = 15$). *STEAM* permuted these factors 10 000 times to derive a threshold for significance. Results were visualised in RStudio. A genome-wide significance threshold of $p\text{-value} < 2.5 \times 10^{-6}$ was deemed significant by *STEAM*.

Results

Global and local ancestry inference

After close inspection of global ancestry proportions generated using ADMIXTURE, the K number of contributing ancestries (the lowest k-value determined through cross-validation) was $K = 3$ for the Xhosa individuals and $K = 5$ for the SAC individuals (Figure 1). This is consistent with previous global ancestry deconvolution results (Chimusa et al., 2014; Choudhury et al., 2021). It is evident that our cohort is a complex, highly admixed group with ancestral contributions from the indigenous KhoeSan (~22 - 30%), Bantu-speaking African (~30 - 72%), European (~5 - 24%), Southeast Asian (~11%) and East Asian (~5%) population groups.

Local ancestry was estimated for all individuals. Admixture between geographically distinct populations creates complex ancestral and admixture-induced LD blocks, which can be visualised using local ancestry karyograms. Figure 2 shows karyograms for three individuals from the merged dataset. It is evident that, despite individuals being from the same population group, each possesses unique patterns of local ancestry arising from differing numbers and lengths of ancestral segments.

Figure 1.

Genome-wide ancestral proportions of all individuals in the merged dataset. Ancestral proportions for each individual are plotted vertically with different colours representing different contributing ancestries.

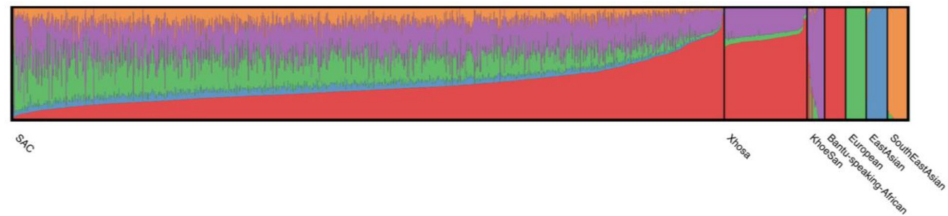
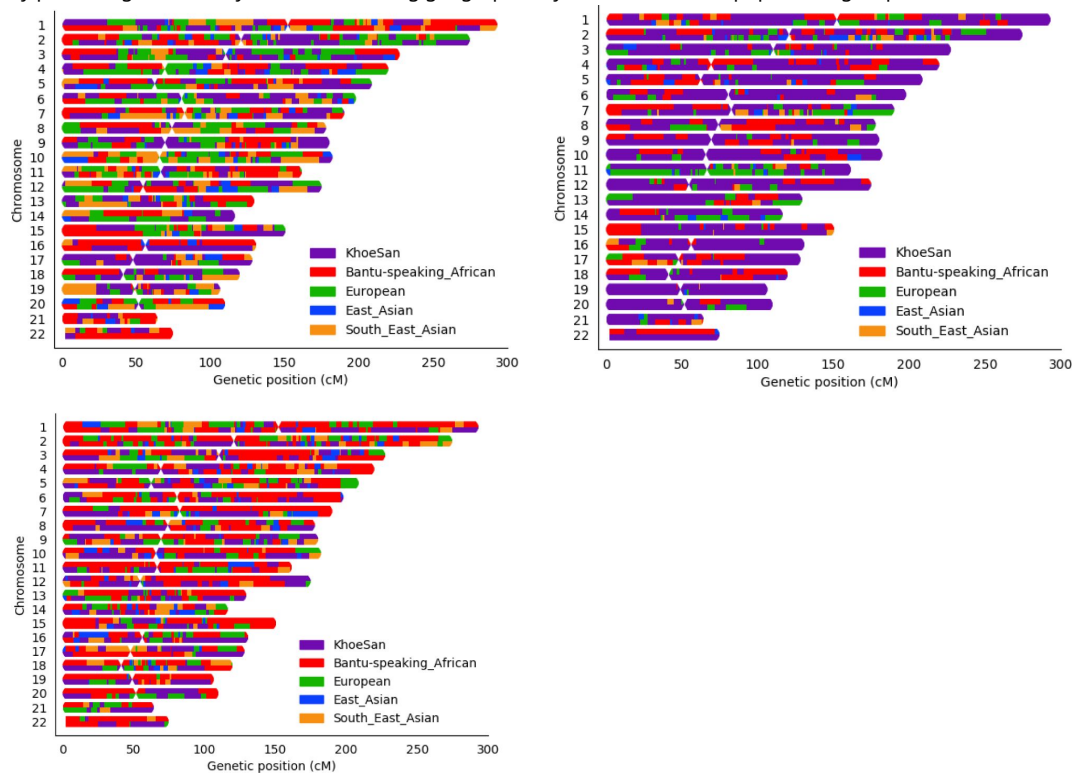


Figure 2.

Local ancestry karyograms of three admixed individuals from the SAC population. Each admixed individual has unique local ancestry patterns generated by admixture among geographically distinct ancestral population groups.



Local ancestry-allelic adjusted analysis

A total of 784 557 autosomal markers (with MAF > 1%) and 1 544 unrelated individuals (952 TB cases and 592 healthy controls) were included in logistic regression models to assess whether any loci and/or ancestries were significantly associated with TB status (whilst adjusting for sex, age, and global ancestry proportions). LAAA models were successfully applied for all five contributing ancestries (KhoeSan, Bantu-speaking African, European, East Asian and Southeast Asian). Only one variant (*rs74828248*) was significantly associated with TB status (p -value < 2.5×10^{-6}) whilst utilising the LAAA model and whilst adjusting for Bantu-speaking African ancestry on chromosome 20 (p -value = 2.272×10^{-6} , OR = 0.316, SE = 0.244) (**Supplementary Figure 1** [↗](#)). No genomic inflation was detected in the QQ-plot for this region (**Supplementary Figure 2** [↗](#)). However, this variant is located in an intergenic region and the link to TB susceptibility is unclear (**Supplementary Figure 3** [↗](#)).

Although no other variants passed the genome-wide significance threshold, multiple lead SNPs were identified. Notably, an appreciable peak was identified in the HLA-II region of chromosome 6 when using the LAAA model and adjusting for KhoeSan ancestry (**Figure 3** [↗](#)). The QQ-plot suggested minimal genomic inflation, which was verified by calculating the genomic inflation factor ($\lambda = 1.05289$) (**Supplementary Figure 4** [↗](#)). The lead variants identified using the LAAA model whilst adjusting for KhoeSan ancestry in this region on chromosome 6 are summarised in **Table 3** [↗](#). The association peak encompasses the *HLA-DPA1/B1* (major histocompatibility complex, class II, DP alpha 1/beta 1) genes (**Figure 4** [↗](#)). It is noteworthy that without the LAAA model, this association peak would not have been observed for this cohort. This highlights the importance of utilising the LAAA model in future association studies when investigating disease susceptibility loci in admixed individuals, such as the SAC population.

The lead variant lies within *COL11A2P1* (collagen type X1 alpha 2 pseudogene 1). *COL11A2P1* is an unprocessed pseudogene ([ENSG00000228688](#)). Unprocessed pseudogenes are seldomly transcribed and translated into functional proteins ([Witek & Mohiuddin, 2024](#) [↗](#)). *HLA-DPB1* and *HLA-DPA1* are the closest functional protein-coding genes to our lead variants.

The lead variant identified in the ITHGC meta-analysis, *rs28383206*, was not present in our genotype or imputed datasets. The ITHGC imputed genotypes using the 1000 Genomes (1000G) reference panel ([Schurz et al., 2024](#) [↗](#)). Variant *rs28383206* has an alternate allele frequency of 11.26% in the African population subgroup within the 1000G dataset (<https://www.ncbi.nlm.nih.gov/snp/rs28383206> [↗](#)). However, *rs28383206* is absent from our in-house whole-genome sequencing (WGS) datasets, which include Bantu-speaking African and KhoeSan individuals. This absence suggests that *rs28383206* might not have been imputed in our datasets using the AGR reference panel, potentially due to its low alternate allele frequency in southern African populations. Our merged dataset contained two variants located within 800 base pairs of *rs28383206*: *rs482205* (6:32576009) and *rs482162* (6:32576019). However, these variants were not significantly associated with TB status in our cohort (**Supplementary Table 1** [↗](#)).

Discussion

The LAAA analysis of host genetic susceptibility to TB, involving 942 TB cases and 592 controls, identified one suggestive association peak adjusting for KhoeSan local ancestry. The association peak identified in this study encompasses the *HLA-DPB1* gene, a highly polymorphic locus, with over 2 000 documented allelic variants ([Robinson et al., 2020](#) [↗](#)). This association is noteworthy given that *HLA-DPB1* alleles have been associated with TB resistance ([Dawkins et al., 2022](#) [↗](#);

Figure 3.

Log transformation of association signals obtained for KhoeSan ancestry whilst using the LAAA model on chromosome 6. The dashed red line represents the significant threshold for admixture mapping calculated with the software STEAM (p -value = 2.5×10^{-6}) and the black solid line represents the genome wide significant threshold (p -value = 5×10^{-8}). The four different models are represented in black (global ancestry only - GAO), blue (local ancestry effect - LAO), orange (ancestry plus allelic effect - APA) and pink (local ancestry adjusted allelic effect - LAAA).

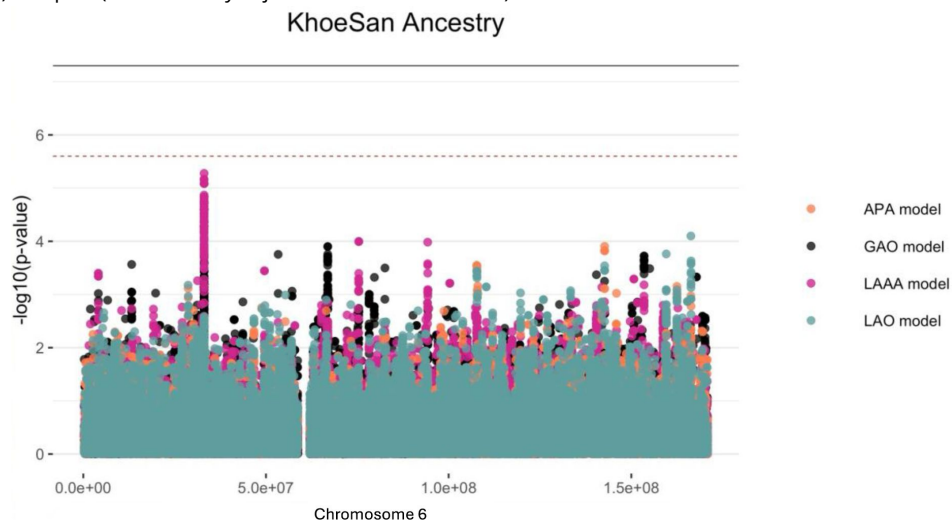
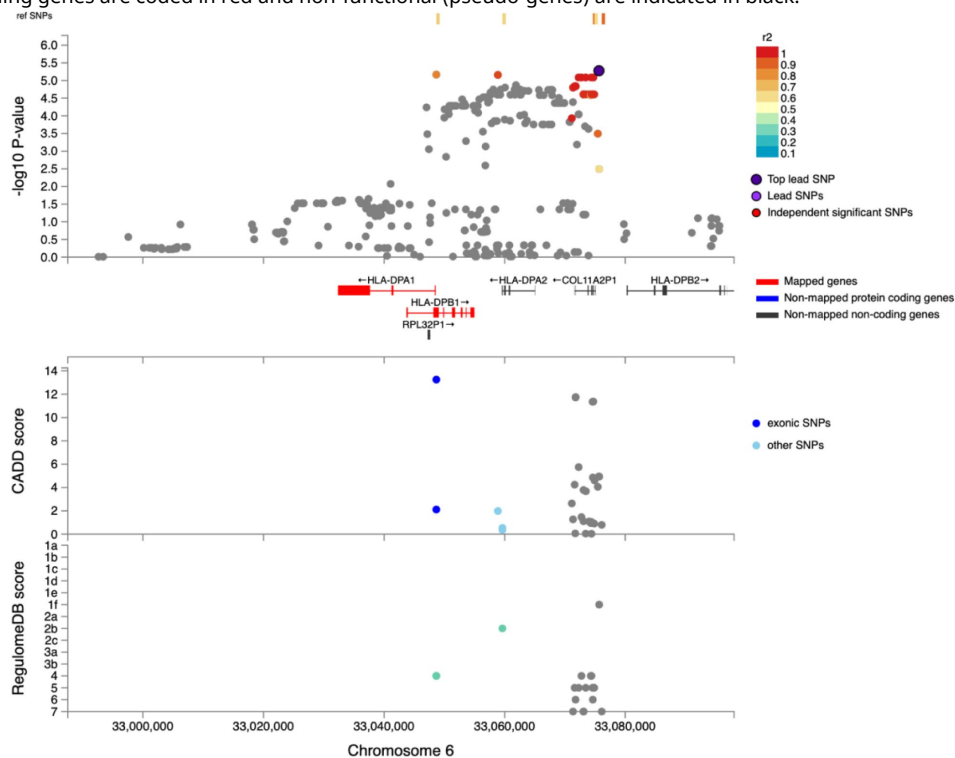


Figure 4.

Regional plot indicating the nearest genes in the region of the lead variant (*rs3117230*) observed on chromosome 6. SNPs in linkage disequilibrium (LD) with the lead variant are coloured red/orange. The lead variant is indicated in purple. Functional protein-coding genes are coded in red and non-functional (pseudo-genes) are indicated in black.



Position	Marker name	Ref	Alt	AltFreq	OR (95% CI)	SE	p-value (x10 ⁻⁶)	Gene	Location	Imputed/typed	INFO score
33075635	rs3117230	A	G	0.370	0.437 (0.306; 0.624)	0.182	5.292	HLA-DPB1	Intergenic	Genotyped	NA
33048661	rs1042151	A	G	0.325	0.437 (0.305; 0.627)	0.184	6.806	HLA-DPB1	Exonic	Imputed	0.992
33058874	rs2179920	C	T	0.369	0.445 (0.313; 0.633)	0.180	6.960	HLA-DPB1	Intergenic	Genotyped	NA
33072266	rs2064478	C	T	0.371	0.447 (0.313; 0.637)	0.181	8.222	HLA-DPB1	Intergenic	Imputed	1
33072729	rs3130210	G	T	0.371	0.447 (0.313; 0.637)	0.181	8.222	HLA-DPB1	Intergenic	Imputed	0.999
33073440	rs2064475	G	A	0.371	0.447 (0.313; 0.637)	0.181	8.222	HLA-DPB1	Intergenic	Imputed	1
33074348	rs3117233	T	C	0.371	0.447 (0.313; 0.637)	0.181	8.222	HLA-DPB1	Intergenic	Imputed	1
33074707	rs3130213	G	A	0.371	0.447 (0.313; 0.637)	0.181	8.222	HLA-DPB1	Intergenic	Imputed	0.970

Ref, reference allele; Alt, alternate allele; AltFreq, alternate allele frequency; OR, odds ratio; SE, standard error

Table 3.

Suggestive associations (p -value < $1e^{-5}$) for the LAAA analysis adjusting for KhoeSan local ancestry on chromosome 6.

Ravikumar et al., 1999 [\[1\]](#); Selvaraj et al., 2008 [\[2\]](#)). The direction of effect the lead variants in our study (**Table 3** [\[3\]](#)) similarly suggest a protective effect against developing active TB. However, variants in *HLA-DPB1* were not identified in the ITHGC meta-analysis.

Population stratification arising from the highly heterogeneous admixed cohorts might have masked this association signal in the African ancestry-specific association analysis. The association peak in the HLA-II region was only captured using the LAAA model whilst adjusting for KhoeSan local ancestry. This underscores the importance of incorporating global and local ancestry in association studies investigating complex multi-way admixed individuals, as the genetic heterogeneity present in admixed individuals (produced as a result of admixture-induced and ancestral LD patterns) may cause association signals to be missed when using traditional association models (Duan et al., 2018 [\[4\]](#); Swart, van Eeden, et al., 2022).

We did not replicate the significant association signal in *HLA-DRB1* identified by the ITHGC. However, the ITHGC also did not replicate this association in their own African ancestry-specific analysis. The significant association, rs28383206, identified by the ITHGC appears to be tagging the *HLA-DQA1**02:1 allele, which is associated with TB in Icelandic and Asian populations (Li et al., 2021 [\[5\]](#); Sveinbjornsson et al., 2016 [\[6\]](#); Zheng et al., 2018 [\[7\]](#)). It is possible that this association signal is specific to non-African populations, but additional research is required to verify this hypothesis. Both our study and the ITHGC independently pinpointed variants associated with TB susceptibility in different genes within the HLA-II locus (**Figure 5** [\[8\]](#)). The HLA-II region spans ~0.8Mb on chromosome 6p21.32 and encompasses the *HLA-DP*, -*DR* and -*DQ* alpha and beta chain genes. The HLA-II complex is the human form of the major histocompatibility complex class II (MHC-II) proteins on the surface of antigen presenting cells, such as monocytes, dendritic cells and macrophages. The innate immune response against *M.tb* involves phagocytosis by alveolar macrophages. In the phagosome, mycobacterial antigens are processed for presentation on MHC-II on the surface of the antigen presenting cell. Previous studies have suggested that *M.tb* interferes with the MHC-II pathway to enhance intracellular persistence and delay activation of the adaptive immune response (Oliveira-Cortez et al., 2016 [\[9\]](#)). For example, *M.tb* can inhibit phagosome maturation and acidification, thereby limiting antigen processing and presentation on MHC-II molecules (Chang et al., 2005 [\[10\]](#)). Given that MHC-II plays an essential role in the adaptive immune response to TB and numerous studies have identified HLA-II variants associated with TB (Cai et al., 2019 [\[11\]](#); Chihab et al., 2023 [\[12\]](#); de Sá et al., 2020 [\[13\]](#); Harishankar et al., 2018 [\[14\]](#); Schurz et al., 2024 [\[15\]](#); Selvaraj et al., 2008 [\[2\]](#)), additional research is required to elucidate the effects of HLA-II variation on TB risk status.

This analysis has a few limitations. First, unlike the ITHGC manuscript, we did not validate our SNP peak in the HLA-II region through fine mapping. Although we initially considered performing HLA imputation and fine-mapping using the HIBAG R package, as described in the ITHGC article (https://hibag.s3.amazonaws.com/hlares_index.html#estimates [\[16\]](#)), the African HIBAG model was trained on genotype data from African American and HapMap YRI populations, which have minimal to no KhoeSan ancestry. Since our association peak likely originates from KhoeSan ancestral haplotype blocks, using an imputation reference panel that includes individuals with KhoeSan ancestry is essential to this analysis. We acknowledge that HLA typing could validate the importance of our lead SNPs in the HLA-II region and support the LAAA model, but this was not feasible due to the absence of a suitable reference panel that includes KhoeSan ancestry. Second, our analysis has a notable case-control imbalance (cases/controls = 1.610). While many studies discuss methods for addressing case-control imbalances with more controls than cases (which can inflate type 1 error rates (Dai et al., 2021 [\[17\]](#); Öztornaci et al., 2023 [\[18\]](#); Zhou et al., 2018 [\[19\]](#)), few address the implications of a large case-to-control ratio like ours (952 cases to 592 controls). To assess the impact of this imbalance, we used the Michigan genetic association study (GAS) power calculator (Skol et al., 2006 [\[20\]](#)). Under an additive disease model with an estimated prevalence of 0.15, a disease allele frequency of 0.3, a genotype relative risk of 1.5, and a default significance level of 7×10^{-6} , we achieved an expected power of approximately 75%. With a balanced sample

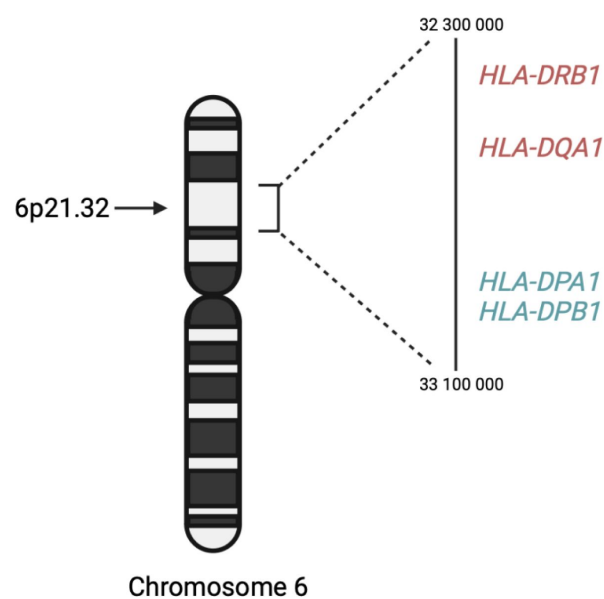


Figure 5.

A schematic diagram the location of HLA-II genes associated with TB susceptibility. Genes in red were identified by the ITHGC. Genes in blue were identified by this study.

size of 950 cases and 950 controls, power would exceed 90%, but it would drop significantly with a smaller balanced cohort of 590 cases and 590 controls. Given these results, we proceeded with our analysis to maximize statistical power despite the case-control imbalance.

In conclusion, application of the LAAA to a highly admixed SAC cohort revealed a suggestive association signal in the HLA-II region associated with protection against TB. Our study builds on the results of the ITHGC by demonstrating an alternative method to identify association signals in cohorts with complex genetic ancestry. This analysis shows the value of including individual global and local ancestry in genetic association analyses. Furthermore, we confirm HLA-II loci associations with TB susceptibility in an admixed South African population and hope that this publication will encourage greater appreciation for the role of the adaptive immune system in TB susceptibility and resistance.

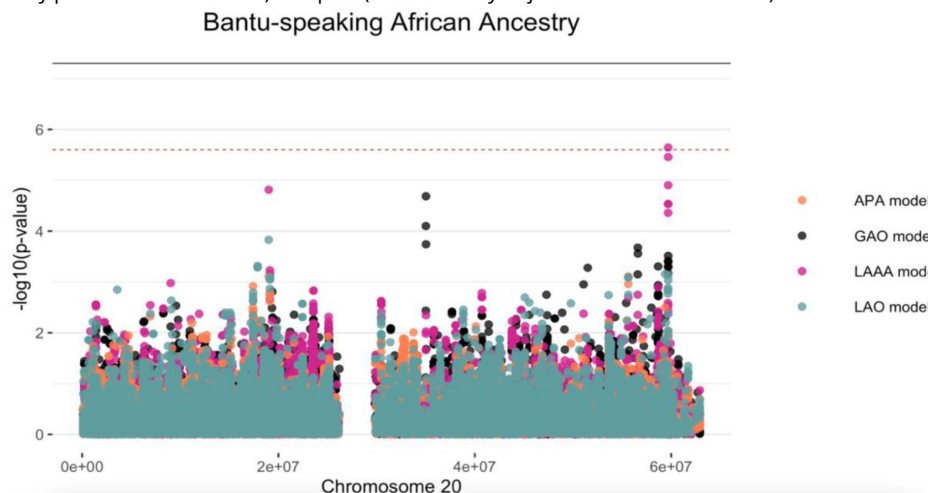
Data Availability

All genetic data used are retrospective datasets and are available through the cited articles.

Supplementary material

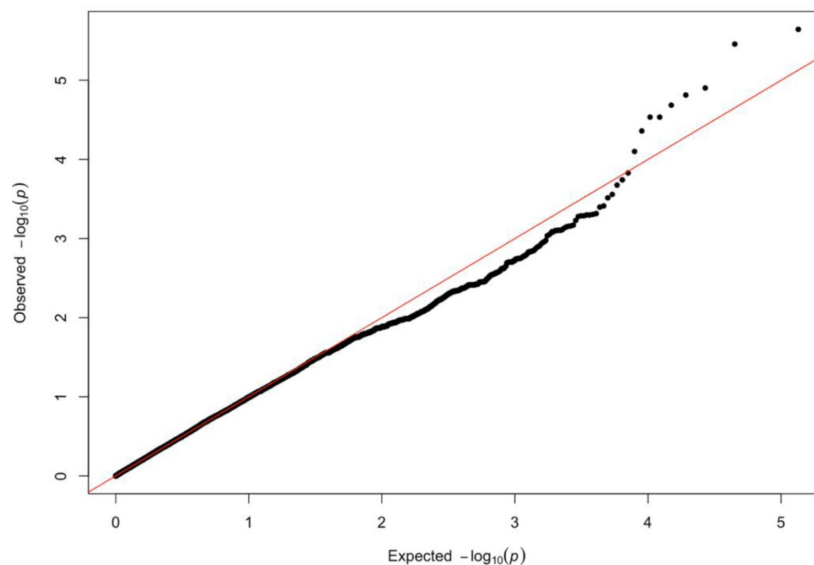
Supplementary Figure 1.

Log transformation of association signals obtained for Bantu-speaking African ancestry whilst using the LAAA model on chromosome 20. The dashed red line represents the significant threshold for admixture mapping calculated with the software STEAM ($p\text{-value} = 2.5 \times 10^{-6}$) and the black solid line represents the genome-wide significant threshold ($p\text{-value} = 5 \times 10^{-8}$). The four different models are represented in black (global ancestry only - GAO), blue (local ancestry effect - LAO), orange (ancestry plus allelic effect - APA) and pink (local ancestry-adjusted allelic effect - LAAA).



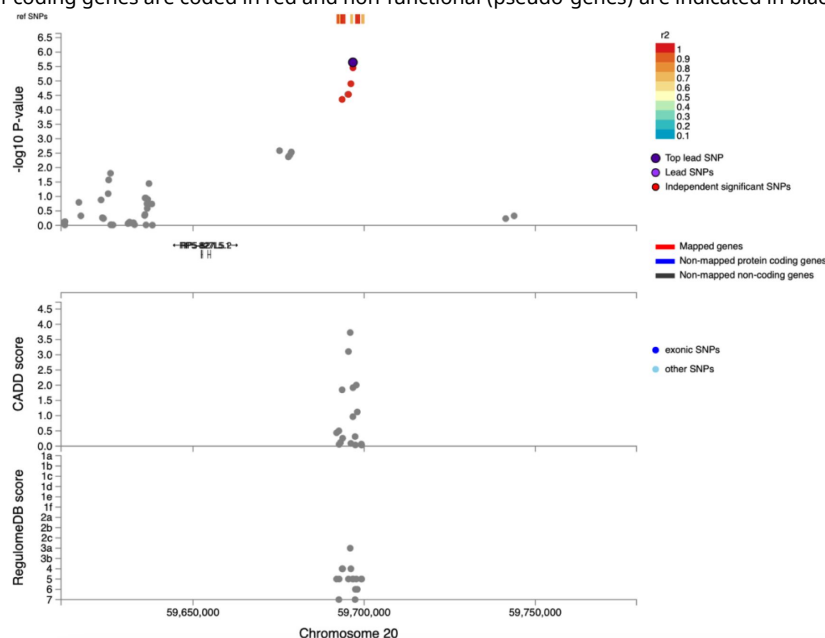
Supplementary Figure 2.

QQ-plot of expected p -values and observed p -values for the association signals obtained for Bantu-speaking African ancestry located on chromosome 20.



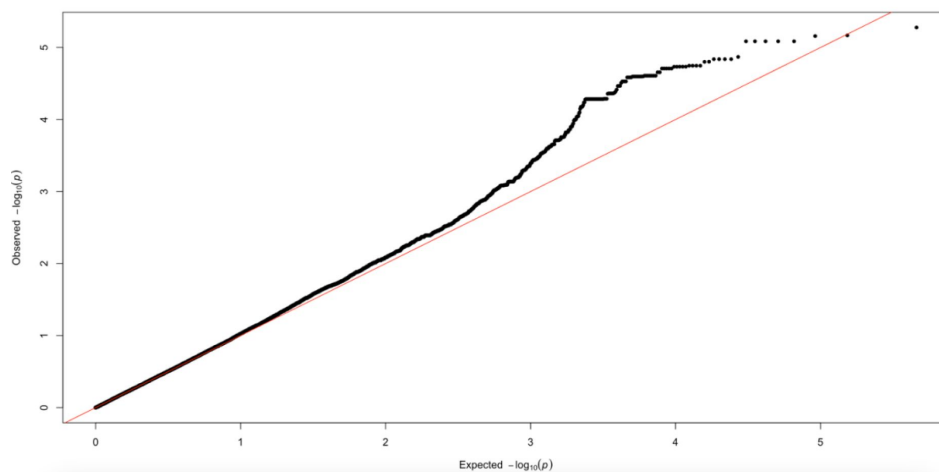
Supplementary Figure 3.

Regional plot indicating the nearest genes in the region of the lead variant (*rs74828248*) observed on chromosome 20. SNPs in linkage disequilibrium (LD) with the lead variant are coloured red/orange. The lead variant is indicated in purple. Functional protein-coding genes are coded in red and non-functional (pseudo-genes) are indicated in black.



Supplementary Figure 4.

QQ-plot of expected p -values and observed p -values for the association signals obtained for Khoisan ancestry located on chromosome 6.



Position	Marker name	Ref	Alt	AltFreq	<i>p</i> -value (KhoeSan local ancestry)	<i>p</i> -value (Bantu-speaking African local ancestry)
32576009	<i>rs482205</i>	T	G	0.322	0.032	0.116
32576019	<i>rs482162</i>	T	C	0.322	0.032	0.116

Supplementary Table 1.

Summary statistics for two variants within 800 base pairs of the ITHGC lead SNP (*rs28383206*) on chromosome 6 for the LAAA analysis adjusting for KhoeSan and Bantu-speaking African local ancestry.

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Additional information

Ethics

Ethics approval was granted by the Health Research Ethics Committee (HREC) of Stellenbosch University, South Africa (project number S22/02/031).

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Reviewer #1 (Public review):

Summary:

The authors aimed to confirm the association between the human leukocyte antigen (HLA)-II region and tuberculosis (TB) susceptibility within admixed African populations. Building upon previous findings from the International Tuberculosis Host Genetics Consortium (ITHGC), this study sought to address the limitations of small sample size and the inclusion of admixed samples by employing the Local Ancestry Allelic Adjusted (LAAA) model, as well as identify TB susceptibility loci in an admixed South African cohort.

Strengths:

The major strengths of this study include the use of multiple TB case-control datasets from diverse South African populations and ADMIXTURE for global ancestry inference.

Weaknesses:

The major weakness of this study include insufficient significant novel discoveries and reliance on cross-validation. The use of existing models did not add value to this study.

Appraisal:

The authors achieved their aims. However, the results still needed to be further validated in the future.

Impact:

The innovative use of the LAAA model and the comprehensive dataset in this study may make contributions to the field of genetic epidemiology.

<https://doi.org/10.7554/eLife.99200.2.sa2>

Reviewer #2 (Public review):

Summary:

This manuscript is about using different analytical approaches to allow ancestry adjustments to GWAS analyses amongst admixed populations. This work is a follow-on from the recently published ITHGC multi-population GWAS (<https://doi.org/10.7554/eLife.84394>), with the focus on the admixed South African populations. Ancestry adjustment models detected a peak of SNPs in the class II HLA DPB1, distinct from the class II HLA DQA1 loci significant in the ITHGC analysis.

Strengths:

Excellent demonstration of GWAS analytical pipelines in highly admixed populations. Particularly the utility of ancestry adjustment to improve study power to detect novel associations. Further confirmation of the importance of the HLA class II locus in genetic susceptibility to TB.

Weaknesses:

Limited novelty compared to the group's previous existing publications and the body of work linking HLA class II alleles with TB susceptibility in South Africa or other African populations. This work includes only ~100 new cases and controls from what has already been published. High resolution HLA typing has detected significant signals in both the DQA1 and DPB1 regions identified by the larger ITHGC and in this GWAS analysis respectively (Chihab L et al. HLA. 2023 Feb; 101(2): 124-137).

Despite the availability of strong methods for imputing HLA from GWAS data (Karnes J et Plos One 2017), the authors did not confirm with HLA typing the importance of their SNP peak in the class II region. This would have supported the importance of this ancestry adjustment versus prior ITHGC analysis.

The populations consider active TB and healthy controls (from high-burden presumed exposed communities) and do not provide QFT or other data to identify latent TB infection.

Important methodological points for clarification and for readers to be aware of when reading this paper:

(1) One of the reasons cited for the lack of African ancestry-specific associations or suggestive peaks in the ITHGC study was the small African sample size. The current association test includes a larger African cohort and yields a near-genome-wide significant threshold in the HLA-DPB1 gene originating from the KhoeSan ancestry. Investigation is needed as to whether the increase in power is due to increased African samples and not necessarily the use of the LAAA model as stated on lines 295 and 296?

Authors response - The Manhattan plot in Figure 3 includes the results for all four models: the traditional GWAS model (GAO), the admixture mapping model (LAO), the ancestry plus allelic (APA) model and the LAAA model. In this figure, it is evident that only the LAAA model identified the association peak on chromosome 6, which lends support the argument that the increase in power is due to the use of the LAAA model and not solely due to the increase in sample size.

Reviewer comment - This data supports the authors conclusions that increase power is related to the LAAA model application rather than simply increase sample size.

(2) In line 256, the number of SNPs included in the LAAA analysis was 784,557 autosomal markers; the number of SNPs after quality control of the imputed dataset was 7,510,051 SNPs (line 142). It is not clear how or why ~90% of the SNPs were removed. This needs clarification.

Authors response:

In our manuscript (line 194), we mention that "...variants with minor allele frequency (MAF) < 1% were removed to improve the stability of the association tests." A large proportion of imputed variants fell below this MAF threshold and were subsequently excluded from this analysis.

Reviewers additional comment: The authors should specify the number of SNPs in the dataset before imputation and indicate what proportion of the 784,557 remaining SNPs were imputed. Providing this information might help the reader better understand the rationale behind the imputation process.

(3) The authors have used the significance threshold estimated by the STEAM p-value < 2.5×10^{-6} in the LAAA analysis. Grinde et al. (2019) implemented their significance threshold estimation approach tailored to admixture mapping (local ancestry (LA) model), where there is a reduction in testing burden. The authors should justify why this threshold would apply to the LAAA model (a joint genotype and ancestry approach).

Authors response: We describe in the methods (line 189 onwards) that the LAAA model is an extension of the APA model. Since the APA model itself simultaneously performs the null global ancestry only model and the local ancestry model (utilised in admixture mapping), we thus considered the use of a threshold tailored to admixture mapping appropriate for the LAAA model.

Reviewers additional comment: While the LAAA model is an extension of the APA model, the authors describe the LAAA test as 'models the combination of the minor allele and the ancestry of the minor allele at a specific locus, along with the effect of this interaction,' thus a joint allele and ancestry effects model. Grinde et al. (2019) proposed the significance threshold estimation approach, STEAM, specifically for the LA approach, which tests for ancestry effects alone and benefits from the reduced testing burden. However, it remains unclear why the authors found it appropriate to apply STEAM to the LAAA model, a joint test for both allele and ancestry effects, which does not benefit from the same reduction in testing burden.

(4) Batch effect screening and correction (line 174) is a quality control check. This section is discussed after global and local ancestry inferences in the methods. Was this QC step conducted after the inferencing? If so, the authors should justify how the removed SNPs due to the batch effect did not affect the global and local ancestry inferences or should order the methods section correctly to avoid confusion.

Authors response: The batch effect correction method utilised a pseudo-case-control comparison which included global ancestry proportions. Thus, batch effect correction was conducted after ancestry inference. We excluded 36 627 SNPs that were believed to have been affected by the batch effect. We have amended line 186 to include the exact number of SNPs

excluded due to batch effect.

The ancestry inference by RFMix utilised the entire merged dataset of 7 510 051 SNPs. Thus, the SNPs removed due to the batch effect make up a very small proportion of the SNPs used to conduct global and local ancestry inferences (less than 0.5%). As a result, we do not believe that the removed SNPs would have significantly affected the global and local ancestry inferences. However, we did conduct global ancestry inference with RFMix on each separate dataset as a sanity check. In the Author response tables 1 and 2, we show the average global ancestry proportions inferred for each separate dataset, the average global ancestry proportions across all datasets and the average global ancestry proportions inferred using the merged dataset. The SAC and Xhosa cohorts are shown in two separate tables due to the different number of contributing ancestral populations to each cohort. The differences between the combined average global ancestry proportions across the separate cohorts does not differ significantly to the global ancestry proportions inferred using the merged dataset.

This is an excellent response and should remain accessible to readers to clarify this issue.

<https://doi.org/10.7554/eLife.99200.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

The authors aimed to confirm the association between the human leukocyte antigen (HLA)-II region and tuberculosis (TB) susceptibility within admixed African populations. Building upon previous findings from the International Tuberculosis Host Genetics Consortium (ITHGC), this study sought to address the limitations of small sample size and the inclusion of admixed samples by employing the Local Ancestry Allelic Adjusted (LAAA) model, as well as identify TB susceptibility loci in an admixed South African cohort.

Strengths:

The major strengths of this study include the use of six TB case-control datasets collected over 30 years from diverse South African populations and ADMIXTURE for global ancestry inference. The former represents comprehensive dataset used in this study and the later ensures accurate determination of ancestral contributions. In addition, the identified association in the HLA-DPB1 gene shows near-genomewide significance, enhancing the credibility of the findings.

Weaknesses:

The major weakness of this study includes insufficient significant discoveries and reliance on crossvalidation. This study only identified one variant significantly associated with TB status, located in an intergenic region with an unclear link to TB susceptibility. Despite identifying multiple lead SNPs, no other variants reached the genome-wide significance threshold, limiting the overall impact of the findings. The absence of an independent validation cohort, with the study relying solely on crossvalidation, is also a major limitation. This approach restricts the ability to independently confirm the findings and evaluate their robustness across different population samples.

Appraisal:

The authors successfully achieved their aims of confirming the association between the HLA-II region and TB susceptibility in admixed African populations. However, the limited number of significant discoveries, reliance on cross-validation, and insufficient discussion of model performance and SNP significance weaken the overall strength of the findings. Despite these limitations, the results support the conclusion that considering local ancestry is crucial in genetic studies of admixed populations.

Impact:

The innovative use of the LAAA model and the comprehensive dataset in this study make substantial contributions to the field of genetic epidemiology.

Reviewer #2 (Public review):

Summary:

This manuscript is about using different analytical approaches to allow ancestry adjustments to GWAS analyses amongst admixed populations. This work is a follow-on from the recently published ITHGC multi-population GWAS (<https://doi.org/10.7554/eLife.84394>), with a focus on the admixed South African populations. Ancestry adjustment models detected a peak of SNPs in the class II HLA DPB1, distinct from the class II HLA DQA1 loci significant in the ITHGC analysis.

Strengths:

Excellent demonstration of GWAS analytical pipelines in highly admixed populations. Further confirmation of the importance of the HLA class II locus in genetic susceptibility to TB.

Weaknesses:

Limited novelty compared to the group's previous existing publications and the body of work linking HLA class II alleles with TB susceptibility in South Africa or other African populations. This work includes only ~100 new cases and controls from what has already been published. High-resolution HLA typing has detected significant signals in both the DQA1 and DPB1 regions identified by the larger ITHGC and in this GWAS analysis respectively (Chihab L et al. HLA. 2023 Feb; 101(2): 124-137). Despite the availability of strong methods for imputing HLA from GWAS data (Karnes J et Plos One 2017), the authors did not confirm with HLA typing the importance of their SNP peak in the class II region. This would have supported the importance of this ancestry adjustment versus prior ITHGC analysis.

The populations consider active TB and healthy controls (from high-burden presumed exposed communities) and do not provide QFT or other data to identify latent TB infection.

Important methodological points for clarification and for readers to be aware of when reading this paper:

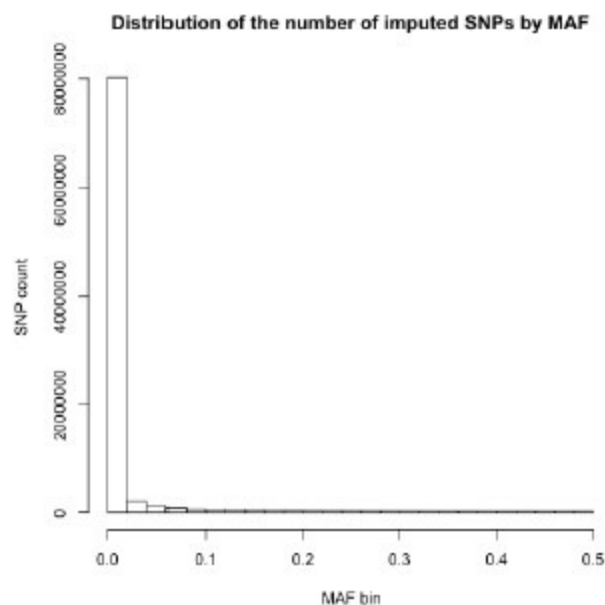
(1) One of the reasons cited for the lack of African ancestry-specific associations or suggestive peaks in the ITHGC study was the small African sample size. The current association test includes a larger African cohort and yields a near-genome-wide significant threshold in the HLA-DPB1 gene originating from the KhoeSan ancestry. The investigation is needed as to whether the increase in power is due to increased African samples and not necessarily the use of the LAAA model as stated on lines 295 and 296?

Thank you for your comment. The Manhattan plot in Figure 3 includes the results for all four models: the traditional GWAS model (GAO), the admixture mapping model (LAO), the ancestry plus allelic (APA) model and the LAAA model. In this figure, it is evident that only the LAAA model identified the association peak on chromosome 6, which lends support the argument that the increase in power is due to the use of the LAAA model and not solely due to the increase in sample size.

(2) In line 256, the number of SNPs included in the LAAA analysis was 784,557 autosomal markers; the number of SNPs after quality control of the imputed dataset was 7,510,051 SNPs (line 142). It is not clear how or why ~90% of the SNPs were removed. This needs clarification.

Thank you for your recommendation. In our manuscript (line 194), we mention that “... variants with minor allele frequency (MAF) < 1% were removed to improve the stability of the association tests.” A large proportion of imputed variants fell below this MAF threshold, and were subsequently excluded from this analysis. Below, we show the number of imputed variants across MAF bins for one of our datasets [RSA(A)] to substantiate this claim:

Author response image 1.



(3) The authors have used the significance threshold estimated by the STEAM p -value < 2.5×10^{-6} in the LAAA analysis. Grinde et al. (2019) implemented their significance threshold estimation approach tailored to admixture mapping (local ancestry (LA) model), where there is a reduction in testing burden. The authors should justify why this threshold would apply to the LAAA model (a joint genotype and ancestry approach).

Thank you for your recommendation. We describe in the methods (line 189 onwards) that the LAAA model is an extension of the APA model. Since the APA model itself simultaneously performs the null global ancestry only model and the local ancestry model (utilised in admixture mapping), we thus considered the use of a threshold tailored to admixture mapping appropriate for the LAAA model.

(4) Batch effect screening and correction (line 174) is a quality control check. This section is discussed after global and local ancestry inferences in the methods. Was this QC step conducted after the inferencing? If so, the authors should justify how the removed SNPs due to the batch effect did not affect the global and local ancestry inferences or should order the methods section correctly to avoid confusion.

Thank you for your comments. The batch effect correction method utilised a pseudo-case-control comparison which included global ancestry proportions. Thus, batch effect correction was conducted after ancestry inference. We excluded 36 627 SNPs that were believed to have been affected by the batch effect. We have amended line 186 to include the exact number of SNPs excluded due to batch effect.

The ancestry inference by RFMix utilised the entire merged dataset of 7 510 051 SNPs. Thus, the SNPs removed due to the batch effect make up a very small proportion of the SNPs used to conduct global and local ancestry inferences (less than 0.5%). As a result, we do not believe that the removed SNPs would have significantly affected the global and local ancestry inferences. However, we did conduct global ancestry inference with RFMix on each separate dataset as a sanity check. In the tables below, we show the average global ancestry proportions inferred for each separate dataset, the average global ancestry proportions across all datasets and the average global ancestry proportions inferred using the merged dataset. The SAC and Xhosa cohorts are shown in two separate tables due to the different number of contributing ancestral populations to each cohort. The differences between the combined average global ancestry proportions across the separate cohorts does not differ significantly to the global ancestry proportions inferred using the merged dataset.

Author response table 1.

Comparison of global ancestry proportions across the separate SAC datasets and the merged cohort.

Cohort	KhoeSan (%)	European (%)	Bantu-speaking African (%)	South East Asian (%)	East Asian (%)
RSA(A)	29.38	23.71	31.04	10.95	4.92
RSA(M)	27.43	25.36	29.64	11.9	5.66
RSA(TANDEM)	27.30	24.28	30.96	11.46	6
RSA(NCTB)	45.92	21.26	24.53	5.28	3
RSA(Worcester)	32.33	21.87	29.16	10.96	5.68
Combined average	32.47	23.30	29.07	10.11	5.05
Merged dataset	29.70	24.10	30.00	10.90	5.20

Author response table 2.

Comparison of global ancestry proportions in the Xhosa dataset and the merged cohort.

Cohort	KhoeSan (%)	European (%)	Bantu-speaking African (%)
RSA(Xhosa)	22.48	4.83	72.68
Merged dataset	22.50	4.80	72.70

Reviewer #1 (Recommendations for the authors):

Suggestions for Improved or Additional Experiments, Data, or Analyses:

(1) It might be beneficial to consider splitting the data into separate discovery and validation cohorts rather than relying solely on cross-validation. This approach could provide a stronger basis for independently confirming the findings.

Thank you for your suggestion. However, we are hesitant to divide our already modest dataset (n=1544) into separate discovery and validation cohorts, as this would reduce the statistical power to detect significant associations.

(2) Clearly stating the process of cross-validation in the methods section and reporting relevant validation statistics, such as accuracy, sensitivity, specificity, and area under the curve (AUC), would provide a more comprehensive assessment of the model's performance.

Thank you for your recommendation. We would like to highlight this article, “GWAS in the southern African context” (1), which evaluated the performance of the LAA model compared to other models in three- and five-way admixed populations. Given the thorough evaluation of the model's performance in that study, we did not find it necessary to reassess its performance in this manuscript.

(3) Analysing racial cohorts separately to see if you can replicate previous results and find significant markers in combined non-African populations that are not evident in African-only samples might be useful.

Thank you for your suggestion. We would like to respectfully note that race is a social construct, and its use as a proxy for genetic ancestry can be problematic (2). In our study, we rather rely on genetic ancestry inferred using ancestry inference software to provide a more accurate representation of our cohort's genetic diversity. Additionally, our cohort consists mostly of a highly admixed population group, with some individuals exhibiting ancestral contributions from up to five different global populations. Therefore, it is not possible to categorize our samples into distinct “Africanonly” or “non-African” groups.

(4) It might be worthwhile to consider using polygenic risk scores (PRS) to combine multiple genetic influences. This approach could help in identifying cumulative genetic effects that are not apparent when examining individual SNPs.

Thank you for your recommendation. While constructing a polygenic risk score (PRS) is beyond the scope of the current study, but an ongoing interest in our group, we recognize its potential value and will consider incorporating this approach in future research endeavours or a separate publication. A recent publication by Majara et al showed that that PRS accuracy is low for all traits and varies across ancestrally and ethnically diverse South African groups (3).

Recommendations for Improving the Writing and Presentation:

Including a more thorough discussion of the methodological limitations, such as the challenges of studying admixed populations and the potential limitations of the LAAA model, would provide a more balanced perspective.

Thank you for your suggestion. To provide a more balanced perspective, we included the limitations of our study in the discussion, from line 429 to line 451.

Minor Corrections to the Text and Figures:

Including all relevant statistics would improve clarity. For example, providing confidence intervals for the odds ratios and discussing any observed trends or outliers would be beneficial.

Thank you for your recommendation. We have added 95% confidence intervals to all odds ratios reported in Table 3. However, beyond the association peak identified in the HL-II region associated with the phenotype, we do not observe any other trends or outliers in our LAAA analysis.

Reviewer #2 (Recommendations for the authors):

Points for improvement:

(1) Related to the different datasets and inclusions in previous publications, it would also be good to better understand the different numbers of cases and controls included across the previous and current analyses, or discussion thereof. For instance, the RSA(M) dataset includes 555/440 cases/controls for this analysis and only 410/405 cases/controls in the ITHGC analysis. Other discrepancies are noted across the other published datasets compared to those included in this analysis, and these always need to be detailed in a supplement or similar to better understand if this could have introduced bias or was in fact correct based on the additional ancestry-related restriction applied.

Thank you for your comments. Table 1 of our manuscript lists number of individuals in the RSA(M) dataset, including related individuals. As described in line 131, related individuals were subsequently excluded during quality control: “Individual datasets were screened for relatedness using KING software (Manichaikul et al., 2010) and individuals up to second degree relatedness were removed.” The ITHGC only reported the number of unrelated individuals included their analyses, which would account for the discrepancies in the reported number of cases and controls.

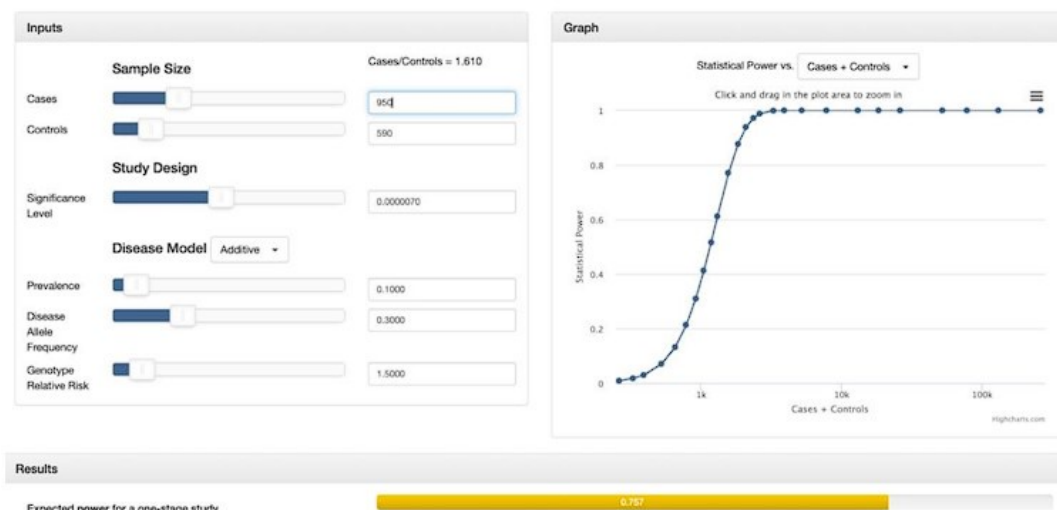
(2) The imbalance between cases and controls in this analysis is quite striking, and it is unusual to have the imbalance favour cases over controls. This contrasts with the ITHGC, where there are substantially more controls. There is no comment on how this could potentially impact this analysis.

Thank you for your comment. We have included a note on our case-control imbalance in the discussion:

“While many studies discuss methods for addressing case-control imbalances with more controls than cases (which can inflate type 1 error rates (Zhou et al. 2018; Dai et al. 2021; Öztornaci et al. 2023), few address the implications of a large case-to-control ratio like ours (952 cases to 592 controls). To assess the impact of this imbalance, we used the Michigan genetic association study (GAS) power calculator (Skol et al. 2006). Under an additive disease model with an estimated prevalence of 0.15, a disease allele frequency of 0.3, a genotype

relative risk of 1.5, and a default significance level of 7×10^{-6} , we achieved an expected power of approximately 75%. With a balanced sample size of 950 cases and 950 controls, power would exceed 90%, but it would drop significantly with a smaller balanced cohort of 590 cases and 590 controls. Given these results, we proceeded with our analysis to maximize statistical power despite the case-control imbalance.”

Author response image 2.



Minor comments

(1) Referencing around key points of TB epidemiology and disease states seems out of date, given recent epidemiology reviews and seminal nature or lancet review articles. Please update.

Thank you for your suggestion. We have included the following recent publications in the introductory paragraph:

Zaidi, S. M. A., Coussens, A. K., Seddon, J. A., Kredo, T., Warner, D., Houben, R. M. G. J., & Esmail, H. (2023). Beyond latent and active tuberculosis: a scoping review of conceptual frameworks. *EClinicalMedicine*, 66, 102332. <https://doi.org/10.1016/j.eclinm.2023.102332>

Menzies, N. A., Swartwood, N., Testa, C., Malyuta, Y., Hill, A. N., Marks, S. M., Cohen, T., & Salomon, J. A. (2021). Time Since Infection and Risks of Future Disease for Individuals with *Mycobacterium tuberculosis* Infection in the United States. *Epidemiology*, 32(1), 70–78. <https://doi.org/10.1097/EDE.0000000000001271>

Cudahy, P. G. T., Wilson, D., & Cohen, T. (2020). Risk factors for recurrent tuberculosis after successful treatment in a high burden setting: a cohort study. *BMC Infectious Diseases*, 20(1), 789. <https://doi.org/10.1186/s12879-020-05515-4>

Escombe, A. R., Ticona, E., Chávez-Pérez, V., Espinoza, M., & Moore, D. A. J. (2019). Improving natural ventilation in hospital waiting and consulting rooms to reduce nosocomial tuberculosis transmission risk in a low resource setting. *BMC Infectious Diseases*, 19(1), 88. <https://doi.org/10.1186/s12879-019-3717-9>

Laghari, M., Sulaiman, S. A. S., Khan, A. H., Talpur, B. A., Bhatti, Z., & Memon, N. (2019). Contact screening and risk factors for TB among the household contact of children with

active TB: a way to find source case and new TB cases. BMC Public Health, 19(1), 1274. <https://doi.org/10.1186/s12889-0197597-0>

Matose, M., Poluta, M., & Douglas, T. S. (2019). Natural ventilation as a means of airborne tuberculosis infection control in minibus taxis. South African Journal of Science, 115(9/10). <https://doi.org/10.17159/sajs.2019/5737>

Smith, M. H., Myrick, J. W., Oyageshio, O., Uren, C., Saayman, J., Boolay, S., van der Westhuizen, L., Werely, C., Möller, M., Henn, B. M., & Reynolds, A. W. (2023). Epidemiological correlates of overweight and obesity in the Northern Cape Province, South Africa. PeerJ, 11, e14723. <https://doi.org/10.7717/peerj.14723>

(2) Lines 46 to 48 appear to have two contradictory statements next to each other. The first says there are numerous GWAS investigating TB susceptibility; the second says there are sparse. Please clarify.

Thank you for bringing this to our attention. We have amended the lines as follows:

“Numerous genome-wide association studies (GWASs) investigating TB susceptibility have been conducted across different population groups. However, findings from these studies often do not replicate across population groups (Möller & Kinnear, 2020; Möller et al., 2018; Uren et al., 2017).”

(3) Add ref in line 69 for two SAC populations.

Thank you for your recommendation. We have included the citation for the ITHGC meta-analysis paper here:

“The authors described possible reasons for the lack of associations, including the smaller sample size compared to the other ancestry-specific meta-analyses, increased genetic diversity within African individuals and population stratification produced by two admixed cohorts from the South African Coloured (SAC) population (Schurz et al. 2024).”

(4) Write out abbreviations the first time they appear (Line 121).

Thank you for your recommendation. We have corrected the sentence as follows:

“Monomorphic sites were removed. Individuals were screened for deviations in Hardy-Weinberg Equilibrium (HWE) for each SNP and sites deviating from the HWE threshold of 10⁻⁵ were removed.”

(5) It would be good in the supplement to see if there is a SNP peak in chromosome 20 with a hit that reached significance in the Bantu-speaking African ancestry.

Thank you for your recommendation. We have included a regional plot for the lead variant identified on chromosome 20 originating from Bantu-speaking African ancestry in the supplementary material (Supplementary Figure 3).

(6) It would be good to mention the p-values of rs28383206 from the ITHGC paper in this cohort for KhoeSan and Bantu-speaking African ancestries.

Thank you for your suggestion. We have included the following paragraph from line 352:

“The lead variant identified in the ITHGC meta-analysis, rs28383206, was not present in our genotype or imputed datasets. The ITHGC imputed genotypes using the 1000 Genomes (1000G) reference panel (4). Variant rs28383206 has an alternate allele frequency of 11.26% in

the African population subgroup within the 1000G dataset (<https://www.ncbi.nlm.nih.gov/snp/rs28383206>). However, *rs28383206* is absent from our in-house whole-genome sequencing (WGS) datasets, which include Bantu-speaking African and KhoeSan individuals. This absence suggests that *rs28383206* might not have been imputed in our datasets using the AGR reference panel, potentially due to its low alternate allele frequency in southern African populations. Our merged dataset contained two variants located within 800 base pairs of *rs28383206*: *rs482205_* (6:32576009) and *rs482162* (6:32576019). However, these variants were not significantly associated with TB status in our cohort (Supplementary Table 1).” Supplementary Table 1 can be found in the supplementary material:

(7) It would improve the readability of the ancestry proportions listed on lines 236 and 237 if these population groups were linked with the corresponding specific population used in Figure 1, as has been done in Table 2.

Thank you for your suggestion. We have amended Figure 1 to include the corresponding population labels mentioned in Table 2.

(8) In line 209, it is not clear why the number of alleles of a specific ancestry at a locus is referred to as a covariate in admixture mapping when the corresponding marginal effect is the parameter of interest.

Thank you for bringing this to our attention. We have amended the description as follows:

“(2) Local ancestry (LA) model:

This model is used in admixture mapping to identify ancestry-specific variants associated with a specific phenotype. The LA model evaluates the number of alleles of a specific ancestry at a locus and includes the corresponding marginal effect as a covariate in association analyses.”

(9) Table 3 would benefit from a column on whether the SNP was genotyped or imputed.

Thank you for your suggestion. We have included a column indicating whether the SNP was genotyped or imputed, as well as an additional column with the INFO score for imputed genotypes.

(10) The authors should remove the print and download icons in Figure 1 on lines 240 and 241.

Thank you for your suggestion. We have amended the figure as requested.

(11) In the quality control, the authors use a more relaxed threshold for missingness in individuals (90%) and genotypes (5%) and have strayed away from the conventional 97%-98%. An explanation of the choice of these thresholds will be helpful to the reader.

Thank you for your suggestion. We aimed to use similar genotype and individual missingness thresholds outlined by the ITHGC meta-analysis (which utilised a threshold of 10% for both genotype and individual missingness) and the previous LAAA analysis paper performed by Swart et al. in 2021. We have amended line 116 for more clarity:

“Individuals with genotype call rates less than 90% and SNPs with more than 5% missingness were removed as described previously (5).”

References

- (1) Swart Y, van Eeden G, Uren C, van der Spuy G, Tromp G, Moller M. GWAS in the southern African context. Cold Spring Harbor Laboratory. 2022;
- (2) Byeon YJJ, Islamaj R, Yeganova L, Wilbur WJ, Lu Z, Brody LC, et al. Evolving use of ancestry, ethnicity, and race in genetics research-A survey spanning seven decades. *Am J Hum Genet.* 2021 Dec 2;108(12):2215–23.
- (3) Majara L, Kalungi A, Koen N, Tsuo K, Wang Y, Gupta R, et al. Low and differential polygenic score generalizability among African populations due largely to genetic diversity. *HGG Adv.* 2023 Apr 13;4(2):100184.
- (4) Schurz H, Naranbhai V, Yates TA, Gilchrist JJ, Parks T, Dodd PJ, et al. Multi-ancestry metaanalysis of host genetic susceptibility to tuberculosis identifies shared genetic architecture. *eLife.* 2024 Jan 15;13.
- (5) Swart Y, Uren C, van Helden PD, Hoal EG, Möller M. Local ancestry adjusted allelic association analysis robustly captures tuberculosis susceptibility loci. *Front Genet.* 2021 Oct 15;12:716558.

<https://doi.org/10.7554/eLife.99200.2.sa0>